

LEE JUNWEI

Robotics Simulation Engineer · AirSim · PX4 · ROS2 · Adversarial RL

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SUMMARY

Robotics simulation engineer building high-fidelity, scalable environments for autonomy and policy validation. Full-stack experience across AirSim, PX4, Ardupilot and ROS2, with deep work on sensor realism, multi-agent infrastructure and zero-copy data pipelines. Research background in adversarial reinforcement learning for stress-testing autonomous systems — bridging simulation infrastructure and ML-driven evaluation.

EXPERIENCE

Simulation Engineer

DSTA · Full-time

Jun 2024 – Present · Singapore · Hybrid

- Extended **AirSim** with Fisheye camera models, visual perturbations, and a ROS2 sidecar — improving sensor timing, synchronization, and noise behavior, and enabling multi-camera rig testing with proper intrinsics/extrinsics calibration.
- Containerized multi-agent simulation stacks (**PX4 + Ardupilot + AirSim**) on Docker, enabling parallel scenario rollouts for autonomy validation at scale.
- Built zero-copy sensor transport (**Iceoryx2** / shared memory) achieving 30 Hz multi-camera publishing — unblocking real-time perception training and HIL testing.
- Designed scenario-driven validation workflows in **Argo** with config-driven test injection to systematically probe autonomy under failure modes, degraded sensing, and edge cases.
- Built observability stack (**Elasticsearch + Grafana**) tracking planner metrics — path length, collisions, travel time — across thousands of simulation runs for regression analysis and policy debugging.

AirSim · PX4 / Ardupilot · ROS2 · Docker · Iceoryx2 · Argo

Research Intern — Adversarial RL

DSTA · Internship

Jan 2023 – Jul 2023 · Singapore · Hybrid

- Researched adversarial reinforcement learning for security and robustness evaluation of autonomous systems.
- Built adversarial RL agents to model intelligent attackers and defenders, stress-testing system behavior under worst-case conditions.
- Conducted feasibility studies applying adversarial RL to threat modeling and robustness evaluation of autonomy stacks.
- Implemented training experiments and supporting tooling in Python (PyTorch).

Python · PyTorch · Reinforcement Learning · Adversarial ML

Software Engineering Intern

Variantz Asia-Pacific

May 2021 – Nov 2021 · Singapore

- Designed and built a functional application prototype using NestJS and PostgreSQL, covering schema design, API layer and integration testing.
- Owned product data dashboards to surface usage and operational insights for engineering and product decisions.

NestJS · PostgreSQL · TypeScript

EDUCATION

BEng, Computer Science

Nanyang Technological University

Jan 2020 – Dec 2024

Relevant: AI, Machine Learning, Software Engineering, Computer Vision, Distributed Systems

KEY ACHIEVEMENTS

◆ 30 Hz Multi-Camera Pipeline

Iceoryx2 zero-copy sensor transport unblocking real-time perception training and HIL testing.

◆ Multi-Agent Drone Sim at Scale

Containerized PX4 / Ardupilot / AirSim stacks for parallel scenario rollouts in autonomy validation.

◆ Sim-to-Real Sensor Realism

Fisheye + visual perturbation extensions to AirSim, enabling domain randomization for perception robustness.

◆ Adversarial RL for Autonomy

Research and production workflows applying adversarial RL to stress-test autonomy under edge-case conditions.

LANGUAGES

English — Native

Chinese — Native

CERTIFICATIONS

BrainHack 2021 Participant

DSTA · Defence AI hackathon

Shopee Code League 2021 — Participant

Shopee · Algorithms & data science

TECHNICAL SKILLS

Simulation & Physics

AirSim, Gazebo, Unreal Engine, PX4 SITL, Ardupilot SITL

Robotics & Autonomy

ROS2, multi-agent systems, sensor modeling, sim-to-real, HIL

ML / RL

PyTorch, adversarial RL, reinforcement learning, domain randomization

Languages

Python, C++, C#, TypeScript

Infra & Observability

Docker, Argo Workflows, Iceoryx2, Elasticsearch, Grafana